

The expression method of Chinese creative elements in animation films based on artificial intelligence technology

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ABSTRACT

This study explores the application of artificial intelligence to enhance the representation and evaluation of Chinese cultural elements in animated films, emphasizing both cultural significance and market potential while redefining intellectual property (IP) value in the industry. A major challenge addressed is the accurate assessment of cultural content, as traditional Back Propagation Neural Networks (BPNs) often suffer from slow convergence and local minima issues. To overcome these limitations, the research proposes an improved GA-BP model, combining BPNN's localized optimization with the global search capabilities of Genetic Algorithms (GA). The paper reviews cultural development theories and examines the status of Chinese and international animation IPs. Experimental results show that the GA-BP model achieves higher accuracy and stability than standard BPNs, closely matching expert evaluations. This validates its effectiveness in supporting intelligent cultural evaluation and creative design in animation. By applying AI techniques to cultural evaluation, the research applies artificial intelligence methods to evaluate and support the structured integration of cultural elements into animated film design, laying a methodological groundwork for innovation in Chinese animated films. It supports cultural sustainability and strengthens national cultural identity through digital storytelling, contributing to both academic inquiry and industry practice.

1. Introduction

1.1. Overview

In 2017, the state issued an outline of the development and reform plan, which pointed out that new development concepts should be incorporated into the process of cultural development and reform, specifically developing the cultural industry. At the 2018 Tencent New Cultural and Creative Ecosystem Conference, the vice president of Tencent Group originally developed the notion of “new cultural creation” in response to the present development trend in the cultural business. This study defines cultural expression as the depiction of traditional values, aesthetics, symbolic themes, and identity-driven narratives in animated films. It represents the visual and thematic manifestation of a nation's cultural heritage. Digital cultural production and Intellectual Property (IP) development may be improved by encouraging a greater variety of topic links, which in turn increases the mutual empowerment of cultural value and industrial value [1]. To encourage the change and upgrading of cultural production techniques, new cultural creativity differs from the previously suggested pan-

entertainment model based on the fan economy as the major body by emphasizing the building of creative IP. Moving from a focus on “pan-entertainment” to one that emphasizes “new cultural creation” is a significant step in fostering the preservation and growth of our cultural traditions while also opening up new avenues for creative expression. With the advent of the modern cultural and creative era, many opportunities for cultural development have emerged one after another, and the concept of IP has also undergone conceptual optimization and upgrading from pan-entertainment to new cultural creation [2]. The original IP incubation process, based on the fan economy and emphasizing industrial benefits, has neglected cultural connotations, a lack of copyright awareness, and IP abuse during the operation of the market. The content rectification of related new media platforms, including Kuaishou, Neihan Duanzi, and Toutiao, also reflects from the side that the construction of “pan-entertainment” IP is no longer the advancement trend of my country's cultural and creative industry. The double standard of “business + culture” has become the center of the new cultural creation industry's growth, as it places more emphasis on the creation of IP cultural value [3]. The advancement of digital technology in the Internet age and the communication environment characterized by mobile internet and media integration have provided technical resource

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Nomenclature		Greek letters	
<i>Abbreviations</i>		α	Learning rate or crossover parameter
AI	Artificial Intelligence	δ	Small mutation variation
BPNN	Back Propagation Neural Network	IoT	Internet of Things
GA	Genetic Algorithm	IP	Intellectual Property
GA-BP	Genetic Algorithm Optimized Back Propagation Neural Network	MSE	Mean Squared Error
<i>Subscripts and superscripts</i>		k	Index for output neuron
i	Index for the neuron in the input or hidden layer	t	Time step or generation in iterative processes
j	Index for input neuron	p	Input vector to the neuron
<i>Symbols</i>		R	Length of chromosome encoding (total number of weights and thresholds)
a_{1i}	Output of the i -th neuron in the first (hidden) layer	r	Count of neurons in the input layer
a_{2k}	Output of the k -th neuron in the output layer	S	Number of hidden neurons
b_{2k}	Bias of the k -th neuron in the second layer	s_1	Count of neurons in the first (hidden) layer
B	Bias parameters in the neural network	s_2	Count of neurons in the second (output) layer
E	Error function	$w_{-}\{ij\}$	Weight from input neuron j to
$f(i)$	Fitness of the i -th individual	W	The set of weights in the neural network
F_1	Activation function of the hidden layer	$w_{-}\{ki\}$	Weight from hidden neuron i to output neuron k
F_2	Activation function used in the second layer	X	Count of chromosomes in the population
M	Count of output neurons	Y	Population size
N	Count of input neurons	θ	The threshold in neuron activation
$p(i)$	Probability of selecting the i th individual	η	positive scalar value

advantages for developing the IP industry in the modern cultural and creative era. In this process, the same IP content is disseminated and presented across different platforms in a cross-media narrative. In the current construction of cultural IP content, such as online novels, movies, online dramas, games, etc., the development of IP by domestic animation films has been particularly prominent in recent years [4]. The innovative creation of the IP of Chinese cultural elements by the new domestic animation films has subverted people's impression of the previous domestic animation films and also opened up the foundation for the development of the IP industry chain of animation series films. But it can't be denied that there are only a handful of successful domestic animation films, and more domestic animation films have been buried in the current film and television market [5]. Simultaneously, issues such as the development of animated film series derivatives, copyright maintenance, and production funding have gradually been exposed. From this aspect, it also shows that the current domestic animation films need to make more efforts and attempts in the construction of the cultural value of IP to obtain both a good harvest in the market and word of mouth and, simultaneously, to continue their IP industry chain [6]. Today, the transformation of the film and television industry from pan-entertainment focusing on popular hits to new cultural and creative creation of "IP clusters" is to explore the balance between the commercialization and creation of film and television works and the balance between product management and artistic work. It is also the changing direction of the development and construction of film and television IP in the new cultural and creative era. Domestic animated films, as a unique genre branch of Chinese films, are gradually being endowed with more cultural connotations and meanings beyond films [7]. With the help of the dissemination of cultural and creative products, the social influence of high-quality IP can be expanded. The fundamental function of cultural and creative product dissemination is to disseminate information about film, cultural, and creative products, that is, to deploy "news" to disseminate news information. Through the dissemination of Chinese film cultural and creative products, the public can learn about the film IP culture integrated with the products. This way offers a profound insight into it and, to a certain extent, promotes the curiosity and sharing desire of the target group and then expands the

social impact of high-quality IP [8]. With the help of the dissemination of cultural and creative products, the economic benefits of the film industry can be improved. Their dissemination can coordinate the link between film, cultural, and creative products and society, including the dissemination of persuasive information. The dissemination of cultural and creative products affects the public's cognition and emotional attitude towards movies. Simultaneously, film, cultural, and creative products can be designed and produced according to the requirements of the public to obtain greater economic benefits. To a certain extent, the dissemination of cultural and creative products inherits the culture of film as a cultural and creative industry; that is, the dissemination of educational information enables Chinese film culture and knowledge and other spiritual properties to be passed on from generation to generation. Film, cultural, and creative products are essentially empowered by content, and the conversion of content into emotion drives the purchase by the public. Distributing films and other creative works has several positive effects, including fostering the film industry's fast economic growth and increasing the industry's financial rewards as well as facilitating the preservation of film culture for future generations [9]. Many excellent domestic animation films have recently attained relatively good results regarding box office and word of mouth, and domestic research on domestic animation films is also on the rise. His research focuses on developing the animation film industry, the narrative structure of films, and the presentation of film images. Nowadays, the consumption situation and environment at home and abroad have undergone great changes, especially for the domestic animation film market. Domestic animated films are in the golden age of development, but good development does not mean that there are fewer problems. Issues such as incomplete IP development, content homogeneity, and rough production have gradually emerged, affecting the current development trend of domestic animation films. Film and television have a bright future thanks to "new cultural creativity," a concept developed in 2018 for expanding the cultural and creative sectors [10]. As a pivotal section of the development of domestic films, domestic animation films must conform to the advancement trend of the film and television industry of the times. While accurately grasping the core of cultural creation, one realizes the rational development and utilization of animation

IP. This paper examines the issues that arise during the IP development and construction processes by situating them within the cultural advancement background of the current “new cultural creation” era and basing its analysis on the process and characteristics of IP development in domestic animation films. Then design the expression method of Chinese creative elements in animated films and use Artificial Intelligence (AI) to evaluate its rationality. According to the specific case analysis, put forward reasonable development opinions to achieve innovation and breakthroughs in the research object.

1.2. Related works

In the field of film and television literature, IP extends to the carrier of content, products, and audience emotions. The essence of IP is market-proven user emotion bearing, or in other words, market-proven user needs in the creative industry. Ye [11] believed that China redefines the concept of IP, which is creative intellectual property with a high focus, great influence, and ability to reproduce and recreate. For the film industry, whether it is IP adaptation or IP development, it has always been a hot topic worthy of study. As a branch of the current development of the film industry, animation films pay special attention to the development and protection of animation IP. Well-known foreign animation film production companies have created many classic cartoon animation film images and a series of classic film works, and the IP industry chain behind them has also enabled their animation film works to achieve the continuation and development of cultural value [12]. Yu et al. [13] took the man at the helm of Studio Ghibli’s commercial field as the research object to explore the commercial success of its animated films. Among them, it is concluded that the studio is constantly innovating and constructing new IPs. With the appeal of a large number of animation IPs, it is bound to form a good commercial return. Sun et al. [14] concluded that the key to the success of animated IP movies is the operation of the IP content itself through the interpretation of the Despicable Me series of animation IP in terms of character building and storyline interpretation. The development of animation films in my country is relatively late. With the breakthrough of “The Lotus Lantern” in 1999, domestic animation films have entered a stage of rapid development. But it was not until around 2008 that I started to focus on the development of animation IP. For the study of domestic animation films, different domestic scholars have done research from different fields. Including creative characteristics, development stages, character modeling, etc., but there is less research on domestic animation film IP. Wang [15] analyzes the current situation of domestic animation film IP development under the Internet boom. And put forward reasonable suggestions for the operation of animation film IP, given the problems faced by IP, such as bubbles, adaptation, and marketing. Mengyue et al. [16] also faced a series of issues in the operation of domestic animation film IP and proposed the national animation movement, the expansion of the big screen, and the fusion of Chinese and Western animation concepts. Ma et al. [17] briefly sorted out the overseas awards of Chinese animation films since the founding of the PRC and reflected the advancement process of my country’s domestic animation films from the side. It also analyzed the reasons for the weak overseas dissemination of my country’s domestic animation films and concluded that the dual principles of “cross-cultural” and “nationalization” should be followed regardless of the adaptation or originality of animation IP. Barra et al. [18] analyze the development stage and current situation of domestic animation IP to find the bottleneck in the development process of domestic animation IP. Then, analyze the reasons for its emergence and explore optimization strategies regarding the lack of innovation, non-standard operations, and protection. This includes focusing on talent training, boosting IP incubation, building brand communication, and formulating evaluation and management strategies for animation IP development. With the development of new technologies such as digital technology and AI, Tencent proposed to build a digital cultural China. In this process, the audience no longer simply buys the stars but pays more

attention to the meaning and value of the text content itself [19]. At the same time, a large number of predatory film and television IP developments, not focusing on the incubation of IP across cultures and platforms, also make IP lose its core connotation. Many failed film and television works and phenomena show that the development trend of pan entertainment that blindly emphasizes economic and industrial benefits cannot represent the progress direction of new business formats in the future. “New cultural and creative” reawakens the value of IP culture, takes high-quality content as the core, and evolves the criteria for judgment. Whether the work conveys positive values, whether it pays attention to the presentation of national cultural ideas, and whether it is a creative form acceptable to global culture. Based on the unchanged core IP concept, it is emphasized that only IP that embodies cultural value is meaningful [20]. Academia has different interpretations and reflections on the brand-new concept of “new cultural creativity.” Zhang [21] upgrades “pan-entertainment” to “new cultural and creative,” which is depicted in the construction of a new IP development model of “culture + industry” and the technology-enabled content production of “content + form.” As a market element of new business, “new cultural and creative” should establish the IP of the new consumption relationship between enterprises and consumers to bring more possibilities for developing the cultural and creative industry. [22] gave their gameplay utilizing the case of the Forbidden City. With the user as the center and the experience as the core, all products and services start from user needs and market needs. Take fashion as the first, and pay attention to the fashion of products and services. Only with cultural creativity as the soul can it gain a competitive advantage. Based on science and technology, relying on the “Internet +” transformation and upgrading. Taking the industry as the carrier will help the creative transformation of excellent traditional culture into new cultural and creative industries. Zeng [23] mentioned that the current “self,” “fashion,” and “community” have become labels of mainstream cultural consumption. Affected by the potential impact of consumption upgrades, the advancement and change trends of modern cultural and creative industries concentrate on content, resource integration, and technology applications. And in this process, “new cultural and creative” changes to cultural consumption ethics, including consumption concepts, consumption patterns, and consumption habits. Rui et al. [24] mentioned that “new cultural creation” has changed the way of narrative, and storytelling has become a new means of cultural communication in the new era. Emphasizing that the key to “new cultural creation” lies in people and the way of communication, paying more attention to the guidance of values and the connection of emotions, and returning to the truth, goodness, and beauty between people. Liu [25] compares the IP concepts in the “pan-entertainment” era and the “new cultural and creative” era and concludes that IP in the “new cultural and creative” era must possess and carry greater cultural value. Realize the cross-media flow of high-quality content, pay attention to reflecting the connotation of national culture, and further strengthen the development and dissemination of high-quality cultural resources. Wan and Ren [26] utilized convolutional neural networks to examine visual expressiveness in animated films, employing Kung Fu Panda as a case study. Their research emphasized the significance of AI and machine learning in improving cinematic aesthetics and personalizing material. Despite attaining a modest accuracy of approximately 57 %, the model exhibited the potential of artificial intelligence to enhance visual creativity and elevate creative quality in animation production. Liu et al. [27] introduced a technique for creating group animations with neural networks enhanced by a Gaussian mutation evolutionary algorithm. Their method enhanced the flexibility and realism of group behavior by automating model creation using animation scene data, surpassing previous rule-based techniques and demonstrating the efficacy of evolutionary optimization in improving animation authenticity and efficiency. Sharma et al. [28] examined the impact of AI on the animation business, highlighting its role in automating conventional procedures and facilitating innovations such as text-to-storyboard production. Their research

underscored the transition from manual to intelligent workflows, accentuating AI's capacity to transform animation production and creative positions. Pardeshi and Mude [29] investigated the role of AI in augmenting animation through the automation of design and rendering processes, hence enhancing realism and efficiency. They also observed some drawbacks, like the effect of automation on animator positions and industry employment.

1.3. Research gaps

Though growing awareness of the production of cultural intellectual property and animated movies, the current body of knowledge clearly shows certain research gaps. First, although many studies have addressed the commercial success and cultural relevance of animation IP, minimal attention has been paid to scientifically assessing the rationality of cultural elements incorporated in animated works. Current methods lack thorough, statistically based means for evaluating the success of cultural expression. Second, especially in closing the distance between technological modeling and cultural interpretation, utilizing AI approaches in the framework of cultural creativity and animation IP remains limited. Earlier works have either concentrated narrowly on industrial strategies or stressed artistic elements, leaving a methodological gap in combining artificial intelligence to improve cultural evaluation. Third, there is a dearth of thorough models for building cultural IP systems that take content personalization, user behavior, and cultural symbolism all under one roof. Furthermore, few studies have investigated optimization techniques to get beyond the practical limitations of conventional neural network models, such as slow convergence and poor generalization, in cultural data analysis. These gaps draw attention to the necessity of multidisciplinary solutions supporting sustainable development in the animation IP sector, combining cultural theory, computational models, and creative practice.

1.4. Contribution

This work significantly contributed to the advancement of cultural intellectual property development and the assessment of animated films. Initially, it introduced an innovative methodological framework that integrates artificial intelligence with cultural content analysis through GA, thereby enhancing the conventional BPNN and resulting in a GA-BP model. This hybrid methodology enhanced the convergence and accuracy of neural networks in assessing the rationality of Chinese creative elements in animated films. The project established a systematic methodology for the creation of cultural intellectual property, incorporating symbolic extraction techniques, a user behavior model, and a cultural IP gene database, thereby producing a scalable and structured framework for cultural representation. This work presents a collection of 1080 instances of cultural components from Chinese animated films, providing a valuable resource for future research and model validation. The GA-BP model exceeded conventional BPNNs in assessing cultural rationality, proving effective as a decision-support tool for enhancing creative property. This project presents a complete multidisciplinary framework to enhance innovation, cultural sustainability, and the promotion of Chinese identity using animation media by mixing computer modeling with cultural elements.

To emphasize the originality and efficacy of the proposed GA-BP model, Table 1 presents a comparative study of pertinent works in the domain of AI-driven animation analysis. Previous studies, such as those by Wan and Ren [26] and Liu et al. [27], concentrated on visual aesthetics or animation model production but failed to adopt a systematic methodology for assessing cultural content and aligning with expert evaluations. Recent advancements in animation automation have prioritized production efficiency, although they have overlooked the necessary interpretive depth for cultural assessment. This work presents the GA-BP model, which specifically aims to evaluate Chinese cultural expression rationally through the integration of global optimization and

Table 1
Comparison Between Related Works and the Proposed GA-BP Model.

Study	Method	Focus Area	Limitations	Proposed Model Contribution
Wan and Ren [26]	Convolutional Neural Network (CNN)	Visual expression in animated films	Moderate accuracy (~57 %); limited to aesthetic visual output	Improves accuracy using GA-BP and targets deeper cultural evaluation
Liu et al. [27]	Genetic Algorithm + Neural Network	Group animation model generation	Focused on animation, motion, and scene modeling	Applies GA-BP to cultural content evaluation, not just visual design
Pardeshi and Mude [29]	AI-based animation automation tools	Rendering and design process automation	Did not address cultural aspects or expert alignment	Integrates cultural symbolism, expert alignment, and interpretability
Proposed GA-BP Model	Genetic Algorithm + Backpropagation NN	Evaluation of Chinese cultural expression	Addresses subjectivity, enhances prediction stability and accuracy	Offers a reproducible, interpretable framework for cultural content evaluation

local learning. This method enhances prediction accuracy while creating a replicable and interpretable framework appropriate for cultural analysis in animated films.

2. Materials and methods

2.1. System construction of cultural IP from the perspective of cultural creation

This study defines a high-quality representation of Chinese culture in animation as the precise, respectful, and contextually suitable incorporation of traditional cultural elements such as historical motifs, philosophical values, symbolic imagery, and narrative themes into the visual and storytelling components of animated films. Evaluation criteria encompass the coherence of cultural symbolism, alignment with existing cultural norms, and concordance with expert evaluations grounded on Chinese heritage.

This paper defines cultural content evaluation as a systematic procedure for examining the authenticity, representational correctness, and coherence of traditional cultural aspects in animation, utilizing expert judgment and algorithmic assessment through the GA-BP model.

- 1) The construction of a cultural IP system from the perspective of “new cultural creation” offers three key benefits: it encourages designers to reinterpret traditional culture through innovative thinking, fostering deeper understanding and promoting cultural inheritance and innovation; it enables developers to better align cultural IP products with market demands and user preferences, enhancing both product quality and user experience while extending IP life cycles; and it strengthens developers' focus on creative content, promoting the protection and expression of diverse cultural elements.
- 2) The “new cultural creation” standpoint of cultural IP system architecture incorporates a total of five elements: a cultural gene database, a user model library, the extraction and transformation of cultural gene symbols, IP incubation paths, and the full chain of the industry. The database of the cultural resources is classified, and the core values are identified through the gene database, while the user groups are defined via the model library. The designers then find the symbolic cultural elements and visualize them as IP products.

Through the incubation paths and industry, these are developed, and the value is transmitted to the consumers, and IP derivatives and added value are supported.

3) Establish a cultural IP gene database. Cultural genes, as a fundamental genetic code, exist in the deep structure of national culture. It is precisely because of the differences in cultural genes that the historical traditions and cultural characteristics of various countries and nations present huge differences. Taking cultural genes as the unit, the core of establishing the cultural IP gene database lies in the construction of the Chinese cultural heritage standard database, the Chinese cultural gene database, and the databases of various cultural institutions. In the “new cultural creation” era, more emphasis is placed on the integration, sharing, and service of cultural resources. Public cultural institutions, cultural production institutions, and university scientific research institutions authorize existing cultural resources. Data collection, storage, and identification are carried out through emerging technologies, and finally, a cultural IP gene database is formed. The cultural IP gene database is used as a cloud service platform to provide integrated, shared, and serviceable cultural resources for the creative product design of cultural production terminals. To realize the carrier conversion and value reconstruction of intellectual property related to cultural content.

4) Build a user model library. In the context of the “new cultural creation” era, designers have not only played the role of solving problems for users and improving user experience. What’s more important is to play the roles of coordinator, observer, and guide in cultural IP design to truly understand users. Explore users’ awareness and cultural needs from a more diverse perspective, and design thinking from the perspective of empathy. From the user’s point of view, the extracted cultural IP symbols are better combined with user behavior and needs to grasp the rules and characteristics of users and better enhance the user’s sense of cultural experience. With the popularity of emerging technologies, more and more companies focus on using AI, big data, IoT, and other technologies to analyze the cognitive characteristics and group characteristics of users. Through data visualization, user models such as values, consumption habits, and lifestyles of different user groups are formed. In this way, a more precise, professional, and three-dimensional user model library is built, as displayed in Fig. 1.

5) Extraction and transformation of cultural gene symbols. The cultural IP symbol itself has the characteristics of storytelling, extensibility,

vivid imagery, and a high degree of recognition. Using film and television dramas, animations, cultural and creative products, etc. as carriers, the cultural IP symbols are visualized, allowing consumers to read and understand different cultural connotations and values. Thus, a cultural IP with distinctive and unique characteristics is formed. “New cultural creation” emphasizes the redesign, induction, and summary of cultural gene symbols with modern design methods, which is the most important part of transforming cultural gene symbols into cultural IP symbols. The cultural gene symbols are extracted from the cultural IP gene database, and cultural producers analyze cultural characteristics and deconstruct cultural semantics and deeply dig into the cultural connotation, aesthetic spirit, and creation concept contained in culture. Through metaphor, abstraction, combination, deconstruction, and other design methods, the cultural gene symbol information is designed and transformed, and the characteristics are condensed, and the value symbols and emotional element symbols with IP traits are extracted. Combined with the symbols and aesthetics of the times, the design and development of cultural resources will be carried out, and finally, a cultural IP with its characteristics will be formed.

2.2. BPNN

2.2.1. Review of BPNN

This study employed a three-layer fully connected BPNN, comprising an input layer with 16 neurons representing extracted cultural features, a hidden layer with 10 neurons, and an output layer with 1 neuron indicating the predicted cultural evaluation score. The sigmoid function is employed as the activation function in the hidden and output layers. This structure was chosen to ensure an adequate level of abstraction while preserving computing efficiency, considering the dataset’s size and complexity. The architecture harmonizes model expressiveness with generalization, rendering it suitable for the subjective and high-dimensional characteristics of cultural content assessment.

BPNN is a multilayer feedforward network trained by error back-propagation. Fig. 2 displays its design.

The learning of BPNN encompasses two processes: Information forward and error backpropagation. BPNN has good self-learning, self-adaptation, self-organization, strong fault tolerance, and nonlinear processing ability. Its application range involves expert systems, pattern recognition, and many other fields. BP has the advantages of small computation and strong parallelism. Although BPNN has the above

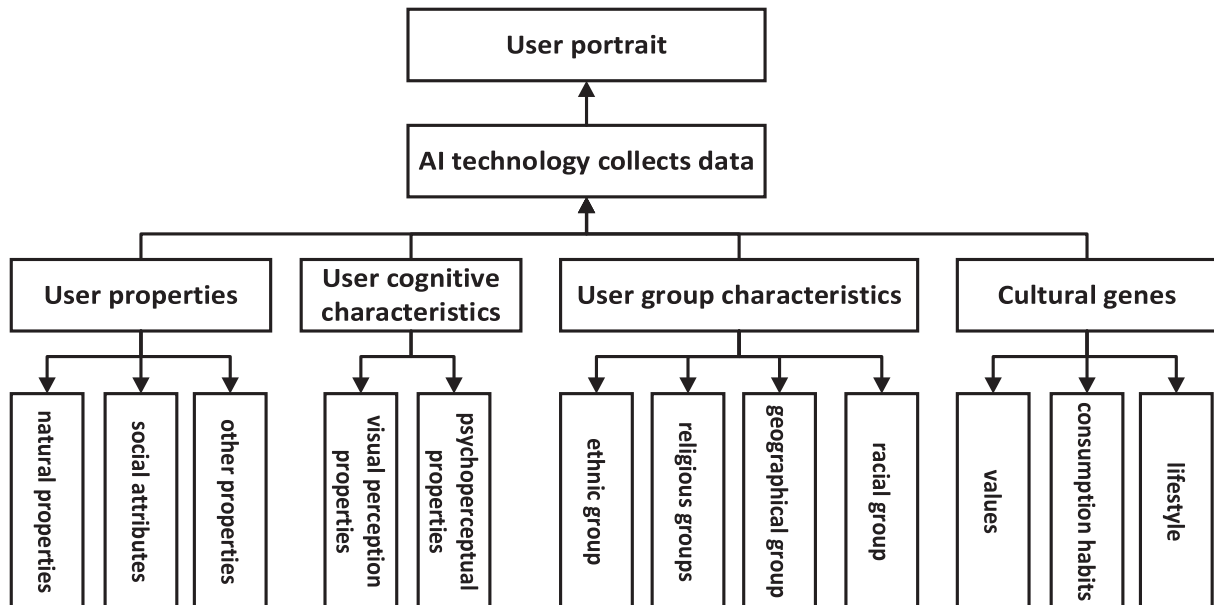


Fig. 1. Construction of user model library from the perspective of new cultural creation.

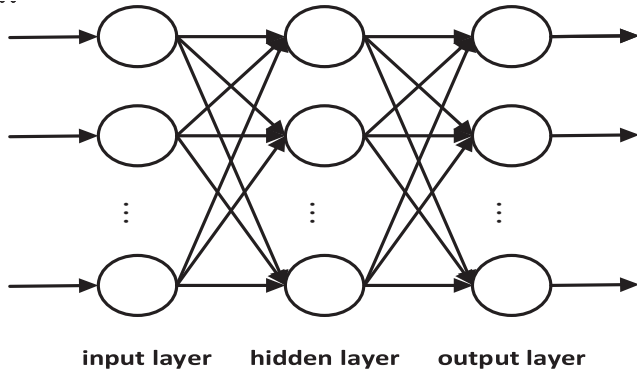


Fig. 2. BPNN basic structure diagram.

significant advantages, it also has its shortcomings. BPNN has a slow convergence speed and low learning efficiency. In the process of optimization, the “sawtooth phenomenon” will occur, and it is easy to fall into a local minimum value. The count of hidden layer nodes is difficult to determine. Therefore, methods to solve these problems need to be explored.

A graphic depiction of the GA-BP learning process is presented in Fig. 3 to enhance clarity and understanding.

Fig. 3 illustrates the comprehensive GA-BP model employed for assessing cultural expression in animated films. The process commences with animation input and feature extraction, followed by a hybrid training framework in which Genetic Algorithms optimize neural network weights before the forward and backward propagation phases. The final product is a quantifiable assessment of cultural content based on expert-aligned grading.

2.2.2. BPNN learning process

Mode forward propagation, error BP, memory training, and learning convergence are the four steps that make up BPNN’s learning process. After the learning process is over, the network obtains an optimal set of weights. This set of optimal weights is the parameters of the evaluation model, and then the network model can be used for evaluation or prediction. BP is essentially a scheme that deploys the squared error sum of the BPNN as the objective function and finds its objective function according to the gradient descent method to reach the minimum value. The learning process of BP includes two stages: forward calculation and error BP. Taking the 3-layer BPNN as an example, the mathematical description of BP is:

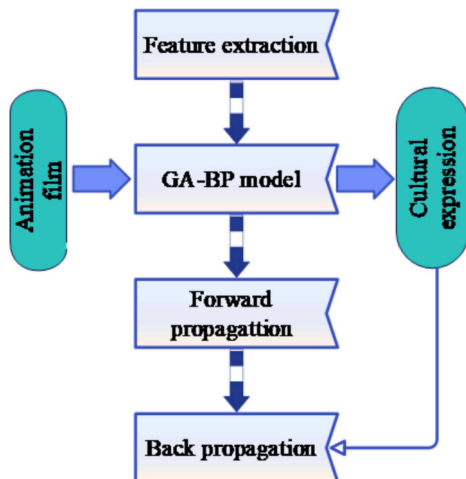


Fig. 3. Flowchart of the GA-BP Learning Process.

1) Forward propagation of information. The output of the i th neuron in the hidden layer is:

$$a_{1i} = F_1 \left[\sum_{j=1}^r w_{1ij} p_j + b_{1i} \right], i = 1, \dots, s_1 \quad (1)$$

where p is the input, r is the count of neurons in the input layer, s_1 displays the count of neurons in the hidden layer, and F_1 is the activation function.

The output of the k th neuron in the output layer is:

$$a_{2k} = F_2 \left[\sum_{i=1}^{s_1} w_{2ki} a_{1i} + b_{2k} \right], k = 1, \dots, s_2 \quad (2)$$

Define the error function as:

$$E(W, B) = \frac{1}{2} \sum_{k=1}^{s_2} (t_k - a_{2k})^2 \quad (3)$$

2) The weight change and the reverse return of the error. For the weights from the i th input to the k th output, there is the following formula:

$$\Delta w_{2ki} = -\eta \frac{\partial E}{\partial w_{2ki}} = -\eta \frac{\partial E}{\partial a_{2k}} \frac{\partial a_{2k}}{\partial w_{2ki}} = \eta \delta_{ki} a_{1i} \quad (4)$$

For the weights from the j th input to the i th output, there is:

$$\Delta w_{1ij} = -\eta \frac{\partial E}{\partial w_{1ij}} = -\eta \frac{\partial E}{\partial a_{2k}} \frac{\partial a_{2k}}{\partial a_{1i}} \frac{\partial a_{1i}}{\partial w_{1ij}} = \eta \delta_{ij} p_j \quad (5)$$

2.2.3. Solution design of BPNN

When designing a BPNN, factors such as the training samples of the network, the count of layers of the network structure, the count of neurons in each layer, the initial weight, and the learning rate should be considered.

- 1) Select input and output sample sets. To test the actual effect of network training, it is generally necessary to sort the sample data first. If the sample data is sufficient, it can be considered to be divided into three groups: training samples, test samples, and evaluation samples. If the sample data is not enough, a small number of samples can be used for testing, and the remaining samples can be used for network training.
- 2) Determine the count of layers of BPNN. Regarding the convergence speed of the network and the avoidance of local minima, the count of hidden layers in the BPNN generally does not exceed two layers. There are different ways to determine the count of hidden layer nodes, which are mainly linked to the requirements of the problem and the count of input and output nodes.
- 3) Initial normalized sample data. Since the unit magnitude of the sample data may not be uniform, it is easy to cause the network to fall into local extreme values. Usually, the training samples need to be normalized to limit the value range of the data.
- 4) Nonlinear fitting is utilized to approximate the target function. An appropriate transfer function and training algorithm are selected. The newff function is called to create a BPNN, where initial weights and thresholds are assigned between the input, hidden, and output layers. Finally, training parameters are set to train the BPNN.
- 5) BPNN model simulation and result analysis. The BPNN performs network simulation by calling the sim function, calculates the correlation coefficient by calling the postreg function, calls the MSE function to calculate the fitting error, and evaluates the output results.

2.2.4. The incorporation of Genetic algorithm and BPNN

This paper presents a three-layer BPNN with 16 input neurons, 10

hidden neurons, and 1 output neuron. The network is initialized with normalized inputs, and the weights and biases are represented as real-valued chromosomes by genetic algorithms. The population size is established at 30, with a crossover probability of 0.7 and a mutation probability of 0.05. The fitness function is defined as the reciprocal of the MSE between anticipated and expert-assessed cultural scores. A rank-based selection technique is employed, along with arithmetic crossover and non-uniform mutation procedures. Following 50 generations, the optimal chromosome is encoded to initiate the BPNN, which is further refined by backpropagation with a learning rate of 0.01.

GA is a scheme with strong global search ability, which can search for the global optimum solution, and has no specific criteria for the objective function; the only issue is that it needs to be computable. Therefore, the GA, which is strong at global search, and the BP algorithm, which is good at local optimization, are coupled to realize the complementary benefits of the two, boost the convergence speed of the scheme, and get the global optimum solution of the issue quickly. The GA optimizes the neural network (NN) and can design an NN with learning, memory, and nonlinear mapping capabilities, as well as global search capabilities. The BPNN is optimized by GA, and the optimized result is called the GA-BP model. GA can not only optimize the connection weights of BPNN but also optimize the network structure and learning rules. But at present, the optimization theory of network design and learning rules is immature and difficult to realize.

This study implements a BPNN using a three-layer design, consisting of 16 input neurons for extracted cultural elements, 10 hidden layer neurons, and one output neuron denoting the cultural evaluation score. Sigmoid activation functions were utilized in both the hidden and output layers. Before training, all input features were standardized to the [0, 1] interval by min-max scaling. The dataset was randomly divided into 70 % for training, 15 % for validation, and 15 % for testing. A parameter tuning process was executed, wherein the quantity of hidden neurons (spanning from 6 to 15) and the learning rate (varying from 0.005 to 0.1) were modified according to validation performance. The ideal configuration was attained using 10 hidden neurons and a learning rate of 0.01. The GA employed real-valued encoding, a population size of 30, a crossover rate of 0.7, and a mutation rate of 0.05. Rank-based selection was utilized, in conjunction with arithmetic crossover and non-uniform mutation techniques, across 50 generations. The highest-performing chromosome was utilized to establish the weights and biases of the BPNN, which were further optimized using standard backpropagation.

The GA-BP model was chosen for this investigation because of its exceptional ability to manage difficult, nonlinear evaluation tasks that typically challenge conventional neural networks. In the realm of cultural content assessment characterized by subjective, multidimensional factors such as symbolism, aesthetics, and narrative coherence, conventional BPNNs are susceptible to local minima and exhibit delayed convergence. The incorporation of Genetic Algorithms facilitates a comprehensive search over the parameter space, enhancing initialization and overall efficacy. GA-BP is particularly adept at assessing Chinese cultural components in animation, where interpretative precision and model stability are essential for reliable outcomes.

This section offers a comprehensive mathematical and procedural delineation of the GA-BP model, encompassing its network architecture, encoding methodology, genetic operators, and fitness assessment. This seeks to guarantee the clarity and reproducibility of the suggested optimization framework.

The optimization process of the GA-BP model can be explicitly articulated as follows.

For a dataset with input features $x \in \mathbb{R}^n$, the forward pass calculates the outputs of the hidden layer [30]:

$$h_i = f\left(\sum_{j=1}^n w_{ji}x_j + b_i\right), i = 1, 2, \dots, S \quad (6)$$

where f is the sigmoid activation function, w_{ij} is the weight between

input node j and hidden node i , and b_i is the bias term for the hidden layer.

The outcome of the output layer is subsequently calculated as:

$$\hat{y} = f\left(\sum_{j=1}^n v_j x_{hj} + c\right) \quad (7)$$

where v_i are the weights connecting the hidden layer to the output layer, and c represents the output bias.

The fitness function for the genetic algorithm is established as the inverse of the Mean Squared Error (MSE).

$$Fitness = \frac{1}{1 + \frac{1}{N} \sum_{k=1}^N (y_k - \hat{y}_k)^2} \quad (8)$$

In GA optimization, the optimal chromosome $\theta^* \in \mathbb{R}^R$ is chosen such that:

$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmax}} Fitness(\theta) \quad (9)$$

where θ encodes all weights and thresholds within the network. The ideal θ^* is utilised to initialize the BPNN before conventional backpropagation fine-tuning.

3. Basic principles

Using GA's chromosome characteristics to encode the weights and thresholds of BPNN is central to the method's success in optimizing these parameters. In this case, the initial population is generated using a specific technique, and the BPNN's fitting error cost function is employed as the GA's fitness function. Individuals are screened through genetic operations, and the best chromosome is chosen as the weight and threshold of BPNN.

3.1. The Main steps of Genetic algorithm (GA) to optimize BPNN

1) Encoding and generation of the initial population. Encodes the connection rights of the network. Binary encoding and floating-point encoding are the most commonly used encoding methods. Among them, binary encoding is simple to operate but not intuitive, and the precision is not high. Floating-point encoding is very intuitive and has high precision. Floating-point number encoding is used because the BPNN's connection weight to the rationality of cultural and creative elements is a floating-point number. As a chromosome of the GA, an array of floating-point numbers is formed from the BPNN's weights and thresholds in a specific order. The rationality evaluation model of cultural and creative elements adopts a three-layer BPNN structure. Assuming that the count of nodes in the input, hidden, and output layers is N , S , and M , accordingly, the length of the code is:

$$R = N \times S + S \times M + S + M \quad (10)$$

The initial group is constructed by haphazardly generating X chromosomes of length R . For the determination of the group number Y , being too large will lead to slow network convergence, and being too small will diminish the accuracy of network training.

2) Determination of fitness function. The R connection weights in the initialized population are allotted to BPNN for forward propagation of the input signal, and the sum of squared errors $E(i)$ between the output value of the network and the expected value is calculated. Set the fitness function as:

$$f(i) = \frac{1}{E(i)} = \frac{1}{\sum_{k=1}^n (y_k - a_k)^2} \quad (11)$$

Combining the GA with the appraisal criteria of the BPNN, the smaller the error sum of squares, the higher the fitness, that is, the better the network performance.

3) Genetic manipulation. Selection is based on a ranking assignment method that takes into account factors like fitness. First, the fitness of each member of the population is determined, and then everyone is ranked according to their fitness. The likelihood of each person getting chosen can be calculated using the ordered list. Based on the distribution principle, the likelihood of being selected is greater for higher fitness values and lower for lower fitness values. The selection method of allocating proportions according to fitness order has the benefit of choosing individuals with high fitness as individuals inherited by the next generation population, and it can also effectively avoid the premature problem of super individuals in the population, which is easily generated when using roulette selection. The crossover operator is the GA's primary function. Crossover in a population creates new individuals, the search space grows over time, and eventually, the goal of global search is met. Crossover is the process of creating two new individuals by crossing the gene chains of two previously chosen individuals with a predetermined probability. The article's crossover operator uses the arithmetic crossover method for floating-point numbers because of their use in the coding. Assuming that x_1 and x_2 in the group are the parent individuals, the offspring x'_1 and x'_2 produced by the parents are:

$$\begin{cases} x'_1 = ax_1 + (1-a)x_2 \\ x'_2 = ax_2 + (1-a)x_1 \end{cases} \quad (12)$$

$$a \in (0, 1)$$

A mutation operator is used to create new people in the population so that the variety of the population is preserved. For certain people, the value of a string in string-structured data may be mutated at random. Here, it is revealed that non-uniform mutation can be used to produce random changes to a gene's value and that the resulting mutated value then takes on the characteristics of the original. All of the loci get the same little tweak. The probability of a mutation ranges from 0.001 to 0.1.

- 4) Procreation of new human beings. New individuals are produced after the first individuals have undergone genetic operations, including crossover and mutation operators. To create a new population, just add some new members to the existing one. Determine whether the new individual's fitness level meets the optimization benchmark or the count of cycles. Proceed to the following phase if so; otherwise, repeat the genetic procedure.
- 5) A BPNN's initial weight generation. The decoding value of the best remaining person in the population is the optimum network connection weight, which is reached when GA achieves the maximum GA or the set index.

3.2. Algorithm flow of GA-BP

The specific stages of GA-BP are:

- 1) According to the problem to be solved, determine the initial structure of the BPNN.
- 2) Pre-train the BPNN for a certain number of times to obtain the range of initial weights and thresholds.
- 3) Float encoding for initial weights and thresholds. Initialize parameters such as population size and evolutionary algebra.
- 4) The role of fitness must be determined. Find the square root of the difference between the BPNN's output and its predicted value, then use the inverse of this value as the fitness function to determine the value of $f(i)$ for individual i .
- 5) Determine whether the individual fitness meets the optimization criteria, and if so, go to step (8). Otherwise, genetic manipulation is performed sequentially.

- 6) Genetic manipulation. Compute the fitness value of each individual in the group and rank them. The probability value of selecting network individuals is displayed in formula (13):

$$p(i) = \frac{f(i)}{\sum_{i=1}^N f(i)} \quad (13)$$

where $i = 1, 2, \dots, N$. It is the count of chromosomes.

The arithmetic crossover method is used for crossover according to the crossover probability. Then use non-uniform mutation to mutate according to the mutation probability.

- 7) Generate new groups. Repeat the process of (3)-(6) to continuously optimize the initially determined weights and thresholds until the optimization criteria are met.
- 8) Decode the optimal solution obtained after GA optimization as the optimal weight and threshold of the BPNN.
- 9) Start training BPNNs. The results of the hidden layer's units and the output units are calculated in advance. Whether the training is complete, you may assess if the accuracy criteria have been fulfilled by calculating the error in the output unit by incrementing it by (10). Otherwise, BPNN training is conducted progressively. Alter the thresholds and weights of the connections from the input to hidden layers and vice versa. Determine whether the training times have been reached, and if the training ends, turn to (10). Otherwise, go back to (9) to repeat the BPNN training.
- 10) BPNN training ends.

4. Experiment and analysis

4.1. Data Sources and preprocessing

The collection consists of 1,080 labeled samples, each depicting cultural content attributes derived from Chinese animated films. Each sample comprises quantitative attributes about conventional symbols, chromatic arrangement, character garments, environment, and thematic elements. The data was partitioned into 70 % for training, 15 % for validation, and 15 % for testing. Before training, all input features were normalized to the [0, 1] range by min-max scaling. The ground truth labels were produced by averaging expert evaluations on a 0-1 scale to denote cultural rationality.

This work utilizes a dataset including 1,080 labeled samples sourced from diverse Chinese cartoon films. Each sample corresponds to a distinct animation scene or fragment and is defined by 16 quantitative parameters that represent essential cultural components. These attributes encompass traditional symbolism, color palettes, attire and costume components, character conduct, narrative framework, incorporation of folklore or historical motifs, and visual aesthetic markers grounded in Chinese culture. All features were standardized to the [0, 1] interval with min-max scaling.

Sixteen input features were derived to encapsulate culturally significant elements in Chinese animated films. These attributes encompass visual aspects (e.g., conventional color schemes, clothing details, and architectural settings), symbolic motifs (e.g., dragons, phoenixes, and mythical allusions), and narrative frameworks (e.g., folklore themes, ethical messages, and character archetypes). Feature extraction was performed via expert-led manual annotation utilizing a set rubric to guarantee uniformity and objectivity. Each attribute received a normalized score ranging from 0 to 1, indicating the extent of cultural representation in each animation clip.

To check the efficacy of the recommended scheme in this exploration, this paper designs a dataset based on the characteristics of Chinese creative elements in animated movies, including 1080 sets of data for training and testing. Before deploying the data, preprocessing is required, and the utilized formula is:

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (14)$$

4.2. Model training

The GA-BP model was created and executed in MATLAB R2021b. All studies were performed on a desktop PC operating Windows 10, equipped with an Intel Core i7-10700 CPU, 16 GB of RAM, and lacking GPU acceleration. Data preparation, model training, and outcome evaluation were conducted utilizing MATLAB's inherent functions for neural networks and bespoke scripts for genetic algorithm operations. The performance measures (MSE, MAE, PCC) were calculated using standard statistical methods executed within the same environment.

- 1) Initialize the BPNN model, and then start network training. The training process of BPNN is displayed in Fig. 4. When the count of training times reaches 35, the target loss is less than 0.001, and the network training ends.
- 2) After setting the parameters of GA-BP, the GA optimizes the weights and thresholds of the BPNN through experiments. The curve diagrams of GA-BP network training loss change and fitness function change are obtained, as displayed in Figs. 5 and 6.

Figs. 5 and 6 Depict the performance dynamics of the ga-bp model throughout the training process. the training loss curve (Fig. 5) exhibits a swift and consistent reduction in error, indicating the model's successful convergence. The fitness function curve (Fig. 6) further substantiates steady enhancement over generations, affirming the Genetic Algorithm's efficacy in attaining superior parameter initialization. These visuals explicitly demonstrate the model's efficacy in enhancing cultural evaluation performance more proficiently than conventional methods.

4.3. Performance test of the optimal model

To validate the correctness of GA-BP in assessing the rationality of cultural and artistic aspects, test data were fed into the two trained models, and the results were compared to expert findings. Table 2 displays the findings collected. The experimental findings display that the output of GA-BP is closer to the expert assessment results, showing that the error of GA-BP is less than that of the BP model. In Table 2, cases 1 to 8 represent eight distinct animation samples, each drawn from different films or scenes with varied cultural content. The GA-BP and BP model outputs for each case reflect the predicted cultural evaluation scores, which are then compared against expert-annotated reference scores. These comparisons illustrate the model's ability to align with expert judgment across diverse cultural expressions.

Table 2 demonstrates the predictive effectiveness of the GA-BP

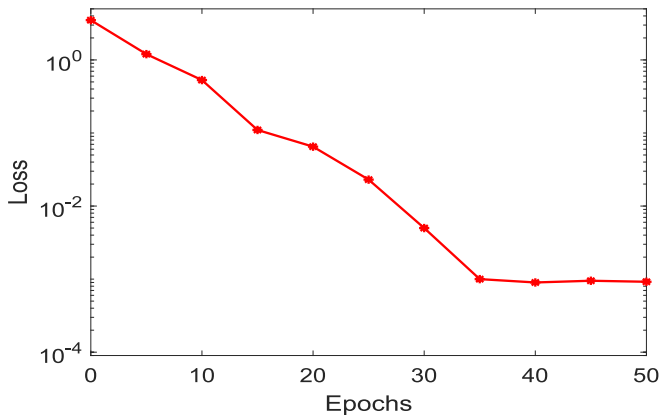


Fig. 4. BPNN training process.

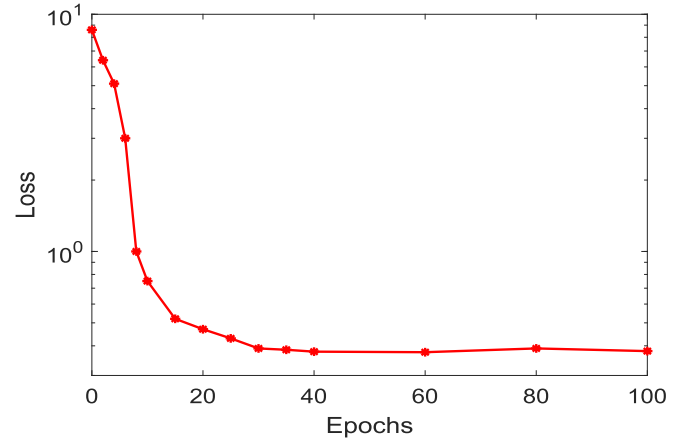


Fig. 5. GA-BP network training loss change curve.

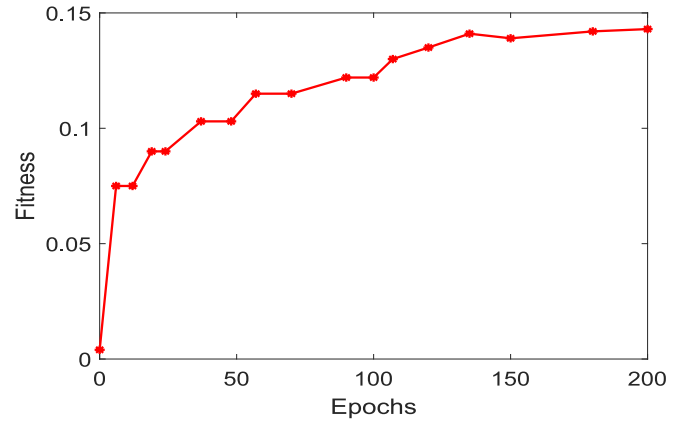


Fig. 6. GA-BP network training fitness function change graph.

model relative to the traditional BPNN and expert assessments across eight exemplary animation samples. Each instance is a unique movie sequence that integrates various cultural components. The GA-BP model regularly generated outcomes that were more closely matched with expert evaluations. MSE, MAE, and PCC were calculated to assess this alignment for both models. The GA-BP model attained a mean squared error MSE of 0.00094, an MAE of 0.021, and a PCC of 0.987, surpassing the BPNN, which produced an MSE of 0.00126, an MAE of 0.029, and a PCC of 0.942. These findings validate the enhanced accuracy, consistency, and predictive efficacy of the GA-BP method in assessing cultural expression in animated content.

Besides visual inspection, evaluation metrics such as MSE, MAE, and PCC were calculated to test the congruence between model predictions and expert evaluations. The GA-BP model attained an MSE of 0.00094 and a PCC of 0.987, surpassing the traditional BPNN, which produced an MSE of 0.00126 and a PCC of 0.942. The results indicate enhanced convergence and more alignment with expert evaluations, validating the resilience of the GA-BP model in cultural content assessment. The GA-BP model presents numerous advantages from a performance standpoint. It markedly enhances convergence velocity and diminishes the probability of entrapment in local minima, a prevalent constraint in conventional BPNNs. Experimental findings include reduced prediction error and enhanced correlation with expert assessments, evidenced by a Pearson Correlation Coefficient of 0.987 and a Mean Squared Error of 0.00094. The results validate that the GA-BP framework offers a more precise and reliable instrument for assessing cultural content in animation.

The GA-BP model's practical application is exemplified by the selection of the Chinese animated feature *White Snake* (2019), notable for

Table 2
Experimental comparison between model output and expert appraisal outcomes.

Number	1	2	3	4	5	6	7	8
GA-BP	0.589	0.633	0.725	0.766	0.906	0.799	0.743	0.839
BP model	0.561	0.639	0.722	0.768	0.910	0.797	0.741	0.843
Expert results	0.588	0.635	0.724	0.766	0.905	0.800	0.745	0.840

its rich use of traditional Chinese cultural themes. A typical setting was examined, yielding 16 cultural attributes, including symbolic motifs like white snake mythology, traditional character garments, architectural components, and narrative themes focused on loyalty and transformation. The trained GA-BP model analyzed these characteristics, yielding a projected cultural evaluation score of 0.897. This output demonstrated a robust correlation with the average expert score of 0.902, thus affirming the model’s efficacy in evaluating cultural representation. The case study demonstrates the model’s interpretive capacity and its relevance in assessing heritage-oriented animation content.

5. Conclusion

This work effectively illustrated how artificial intelligence, more especially a GA-BPNN scheme, could be included in the assessment of Chinese creative elements in animated movies. The GA-BP model attained better convergence speed and evaluation accuracy by resolving constraints in conventional BPNs, such as slow convergence and sensitivity to local minima. With greater alignment to expert evaluations and lower evaluation error, experimental comparisons between model outputs and expert assessments found that GA-BP routinely beat the standard BP model. The approach demonstrated strong flexibility for application in cultural content evaluation, as well as improved the rationality analysis of cultural representation. Furthermore, the work-built a thorough cultural IP system combining symbolic transformation, user behavior modeling, and cultural gene database integration. This framework allowed for a methodical, scalable development and assessment of animation IPs anchored in traditional Chinese culture. The value of the model as a decision-support tool for both creative design and cultural sustainability was validated by its ability to find, organize, and validate cultural materials. These findings help to greatly advance digital cultural production by providing a consistent approach to link technology with cultural expression in animation and therefore promote the strategic growth of globally relevant, culturally rich animated content.

This research is significant for its transdisciplinary contributions to cultural studies and artificial intelligence. This study integrates a GA-BP model with a systematic framework for evaluating cultural content, addressing a significant gap in analyzing the rationality and authenticity of traditional aspects in animation. This method not only improves the accuracy of cultural analysis but also promotes the sustainable development of culturally embedded intellectual property in the animation sector, providing significant insights for both scholarly study and creative output.

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Fengtian Shao: Writing – original draft, Supervision, Project

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Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.entcom.2025.101015>.

Data availability

Data will be made available on request.

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